**AGA KHAN UNIVERSITY EXAMINATION BOARD**

**HIGHER SECONDARY SCHOOL CERTIFICATE**

**CLASS XI EXAMINATION**

**APRIL/ MAY 2018**

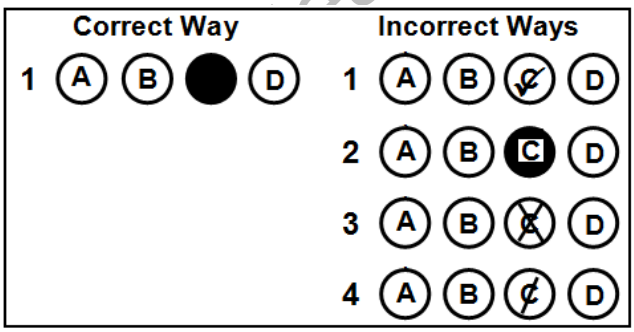
**Chemistry Paper I**

**Time: 40 minutes Marks: 30**

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**INSTRUCTIONS**

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.




**Candidate's Signature**

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The number of moles of $Ca(OH)_2$ that completely reacts with 50 g of $NH_4Cl$ to produce 15.87 g of $NH_3$ gas in the given reaction is

$$2NH_4Cl_{(s)} + Ca(OH)_{2(s)} \longrightarrow CaCl_{2(s)} + 2NH_{3(g)} + 2H_2O_{(l)}$$

(**Note**: Atomic mass of C = 12 amu, H = 1 amu, N = 14 amu, Cl = 35.5 amu and Ca = 40 amu)

A. 0.467
B. 0.934
C. 1.868
D. 3.736

2. Methane burns exothermically in the presence of free oxygen at STP as shown in the chemical equation below.

$$CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(l)}$$

If 25 moles of $CH_4$ combust to give equal moles of $CO_2$, then the volume of $CO_2$ formed will be

A. 0.897 $dm^3$
B. 22.414 $dm^3$
C. 25.00 $dm^3$
D. 560.35 $dm^3$

3. Sodium metal vigorously reacts with water to form sodium hydroxide and hydrogen gas as shown in the given equation.

$$2Na_{(s)} + 2H_2O_{(l)} \longrightarrow 2NaOH_{(aq)} + H_{2(g)}$$

An experiment under controlled environment in a laboratory gives 80% yield of sodium hydroxide on reacting 2 moles of sodium metal with 2 moles of water. How much actual yield of sodium hydroxide in grams is obtained in the given reaction?

A. 32 g
B. 50 g
C. 64 g
D. 100 g

4. In the discharge tube, the original glow disappears when the pressure inside the tube is reduced to

A. 0.01 mm Hg
B. 0.5 cm Hg
C. 1 mm Hg
D. 1 cm Hg

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5. When rapidly moving electrons collide with heavy metal anode in the discharge tube, it produces

   A. X-rays.
   B. gamma rays.
   C. cathode rays.
   D. positive rays.

6. Visible light is NOT used to determine the position of electron because the wavelength of visible light is millions of times larger as compared to the

   A. energy of electron.
   B. velocity of electron.
   C. diameter of electron.
   D. momentum of electron.

7. Mixing of acetic acid with sodium acetate will give buffer with pH

   I. < 7.0
   II. > 7.0
   III. = 7.0

   A. I only.
   B. II only.
   C. III only.
   D. II and III.

8. A basic buffer is prepared by mixing a weak base and a salt of

   A. weak acid with a strong acid.
   B. weak base with a strong acid.
   C. weak acid with a strong base.
   D. weak base with a strong base.

9. If $CH_3COONa$ is added in excess to $CH_3COOH$ solution, then the

   A. pH of solution will increase.
   B. pH of solution will decrease.
   C. dissociation of $CH_3COOH$ will increase.
   D. dissociation of $CH_3COONa$ will decrease.

10. In which of the following molecules the highlighted carbon atom contains two unhybridised p-orbitals?

    A. **C**$H_4$
    B. **C**H≡CH
    C. **C**$H_3$–$CH_3$
    D. **C**$H_2$=$CH_2$

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11. Which of the following bonds between two atoms has the HIGHEST bond length?

    A. B–F
    B. B–Cl
    C. Si–H
    D. Si–F

12. Ionic compounds are unable to show isomerism because in ionic compounds the bonds are

    A. rigid and directional.
    B. non-rigid and directional.
    C. rigid and non-directional.
    D. non-rigid and non-directional.

13. The density of $CH_{4(g)}$ at 17°C and at 1 atmospheric pressure is

    (**Note**: Atomic mass of C = 12 amu, H = 1 amu, and the value of R = 0.0821 $dm^3$ atm $K^{-1}$ $mol^{-1}$)

    A. 0.67 $g/dm^3$
    B. 11.46 $g/dm^3$
    C. 22.32 $g/dm^3$
    D. 380.94 $g/dm^3$

14. According to Charles's law, if a gas is warmed by 1°C at constant pressure, it will expand

    A. twice to its original volume at 0°C.
    B. twice to its original volume at 100°C.
    C. to 1/273 of its original volume at 0°C.
    D. to 1/273 of its original volume at 100°C.

15. Which of the following is FALSE about vacuum distillation?

    A. It requires less fuel.
    B. It requires low temperature.
    C. It reduces the time for distillation.
    D. It allows decomposition of compound.

16. Liquid crystals have wide practical application due to their optical and electrical properties.

    Which of the following is INCORRECT about the use of liquid crystals?

    A. They can be used as temperature sensors.
    B. They are used in the dial of analogue watches.
    C. They are used in skin thermography to detect blockage of veins.
    D. They can be used to find the point of potential failure in electrical circuits.

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17. Graphite exists in the form of layers. It is a conductor parallel to the layers but insulator across the layers. This property of graphite is known as

A. symmetry.
B. anisotropy.
C. isomorphism.
D. polymorphism.

18. In ionic solids, ions are held together by strong electrostatic force of attraction. However, ionic crystals are highly brittle because they

A. have low density.
B. have loose packing of ions.
C. exist as neutral independent molecules.
D. contain opposite ions in parallel layers.

19. The equilibrium constant ($K_C$) for the decomposition of HF at 2000°C is $10^{-13}$ which indicates that it has

| | **Stability** | **Rate of Decomposition** |
|---|---|---|
| A | High | Low |
| B | Low | High |
| C | High | High |
| D | Low | Low |

20. Consider the given reaction at equilibrium.

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

Which of the following would be the result if the concentration of oxygen is increased?

| | **$[SO_2]$** | **$[SO_3]$** | **Equilibrium shift** |
|---|---|---|---|
| A | Increased | Decreased | Right |
| B | Decreased | Increased | Right |
| C | Increased | Decreased | Left |
| D | Decreased | Increased | Left |

21. An activated complex is a combination of all the atoms involved in the reaction. It

A. is a short lived species.
B. is a highly stable species.
C. slowly decomposes into products.
D. involves atoms having minimum energy.

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22. Enzymes are proteins that catalyse organic reactions in the living cells.

The enzyme **X** involved in the given reaction is

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\mathbf{X}} C_6H_{12}O_6 + C_6H_{12}O_6$$

A. lipase.
B. urease.
C. zymase.
D. invertase.

23. A solution is prepared by mixing three different components, **A**, **B** and **C**. If the number of moles of each component is $n_A = 4$ moles, $n_B = 6$ moles, $n_C = 10$ moles, then the mole percent of each component will be

| | **Mole Percent** | | |
|---|---|---|---|
| | **A** | **B** | **C** |
| A | 20 | 30 | 50 |
| B | 0.2 | 0.3 | 0.5 |
| C | 0.4 | 0.6 | 0.10 |
| D | 40 | 60 | 100 |

24. When a non-volatile solute is added to a pure solvent, the

A. boiling point of the solution decreases.
B. freezing point of the solution increases.
C. vapour pressure of the solvent increases.
D. rate of evaporation of the solvent decreases.

25. An example of completely miscible liquids is

A. aniline and water.
B. aniline and hexane.
C. cyclohexane and water.
D. benzene and cyclohexane.

26. In thermochemistry, a liquid whose boiling point is to be determined is considered as a/ an

A. system.
B. final state.
C. initial state.
D. surrounding.

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27. Which of the following is the CORRECT formula for calculating the amount of heat evolved during a neutralisation reaction in a glass calorimeter?

A. Quantity of heat = mass × heat capacity × temperature
B. Quantity of heat = mass × specific heat capacity × temperature
C. Quantity of heat = mass × specific heat capacity × change in temperature
D. Quantity of heat = mass × specific latent heat of fusion × change in temperature

28. Iron is extracted from its ore, haematite, in a blast furnace which involves all of the following chemical reactions. In which of these reactions is carbon reduced?

A. $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$
B. $CO_{2(g)} + C_{(s)} \rightarrow 2CO_{(g)}$
C. $CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$
D. $Fe_2O_{3(s)} + 3CO_{(g)} \rightarrow 2Fe_{(s)} + 3CO_{2(g)}$

29. Which of the following statements best defines standard electrode potential?

A. The potential set up when an electrode undergoes reduction.
B. The potential set up when an electrode undergoes oxidation.
C. The potential of an electrode in 1M solution at 25°C and 1atm.
D. The potential of an electrode in 1M solution at 100°C and 1atm.

30. The reaction between $Cr_2O_7^{-2}$ and $H_2SO_3$ gives the following half reactions.

Reduction half reaction: $Cr_2O_7^{-2} \rightarrow Cr^{+3}$

Oxidation half reaction: $H_2SO_3 \rightarrow HSO_4^-$

The equation which represents the balanced reduction half reaction is

A. $Cr_2O_7^{-2} + 14H^+ + 3e^- \rightarrow Cr^{+3} + 7H_2O$
B. $Cr_2O_7^{-2} + 14H^+ + 6e^- \rightarrow Cr^{+3} + 7H_2O$
C. $Cr_2O_7^{-2} + 14H^+ + 3e^- \rightarrow 2Cr^{+3} + 7H_2O$
D. $Cr_2O_7^{-2} + 14H^+ + 6e^- \rightarrow 2Cr^{+3} + 7H_2O$

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END OF PAPER

# Please use this page for rough work

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